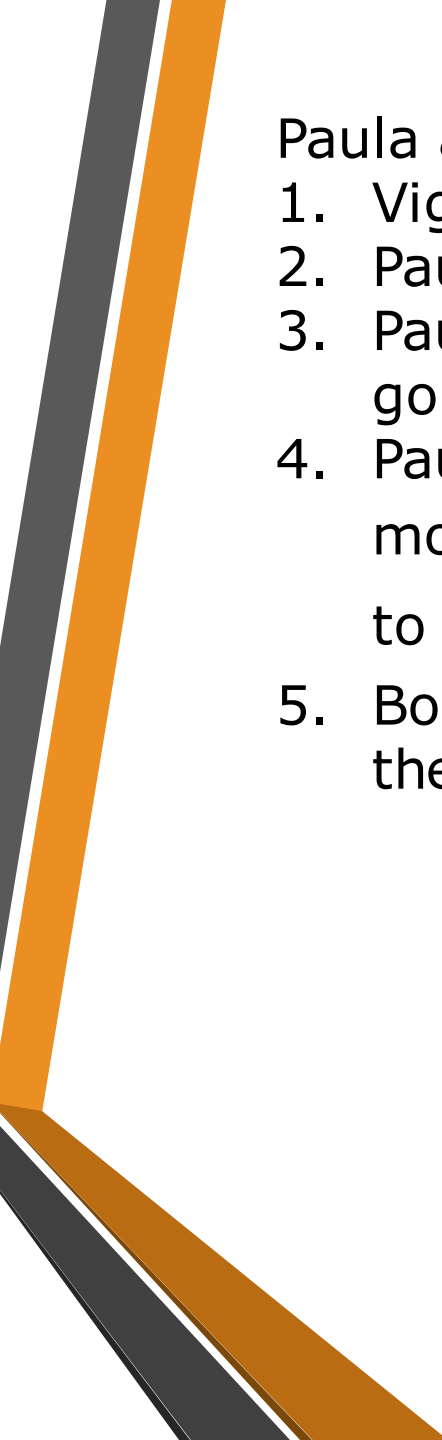




Bayes-Nash Equilibria

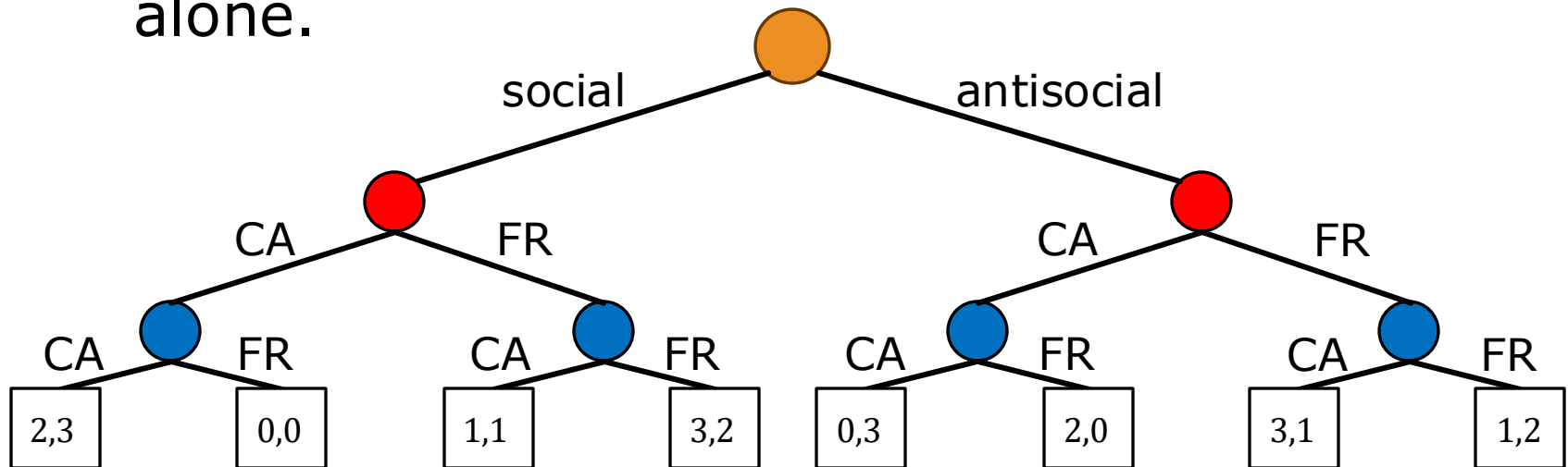


Paula and Vignesh want to watch a movie together.

1. Vignesh prefers Flight Risk to Captain America.
2. Paula prefers Captain America to Flight Risk.
3. Paula prefers to watch a movie with Vignesh to going alone. Vignesh knows this.
4. Paula believes that Vignesh wants to watch a movie with her with prob. $\frac{1}{2}$, and that he prefers to go alone with prob. $\frac{1}{2}$.
5. Both buy tickets without communicating with the other.

Paula gets +1 utility from watching Captain America, and +2 utility from going to the same movie as Vignesh.

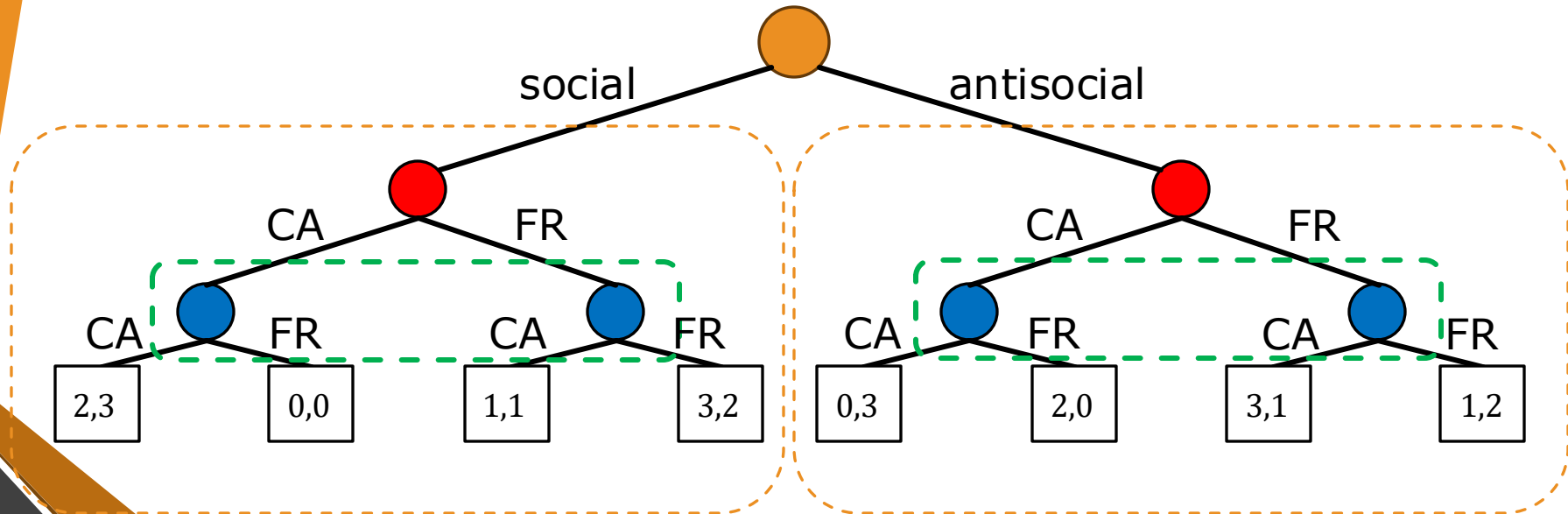
Vignesh gets +1 utility from watching Flight Risk, and +2 utility from going to the same movie as Paula if he wants to go with her, otherwise he gets +2 utility from going alone.



Paula does not know what movie Vignesh is going to, nor does she know whether he's feeling social or antisocial.

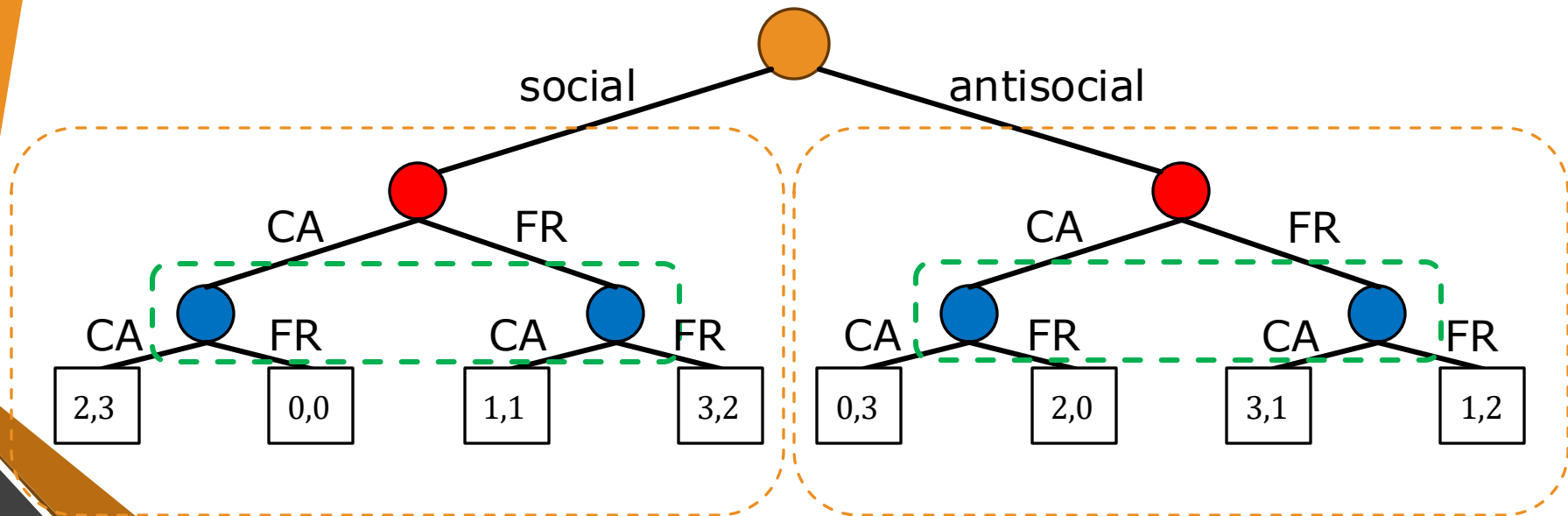
Her **prior** is that he's social with prob. $\frac{1}{2}$.

Vignesh knows how he's feeling, he has **more information**.



Vignesh's strategy can be conditional on being social (s) or antisocial (a).

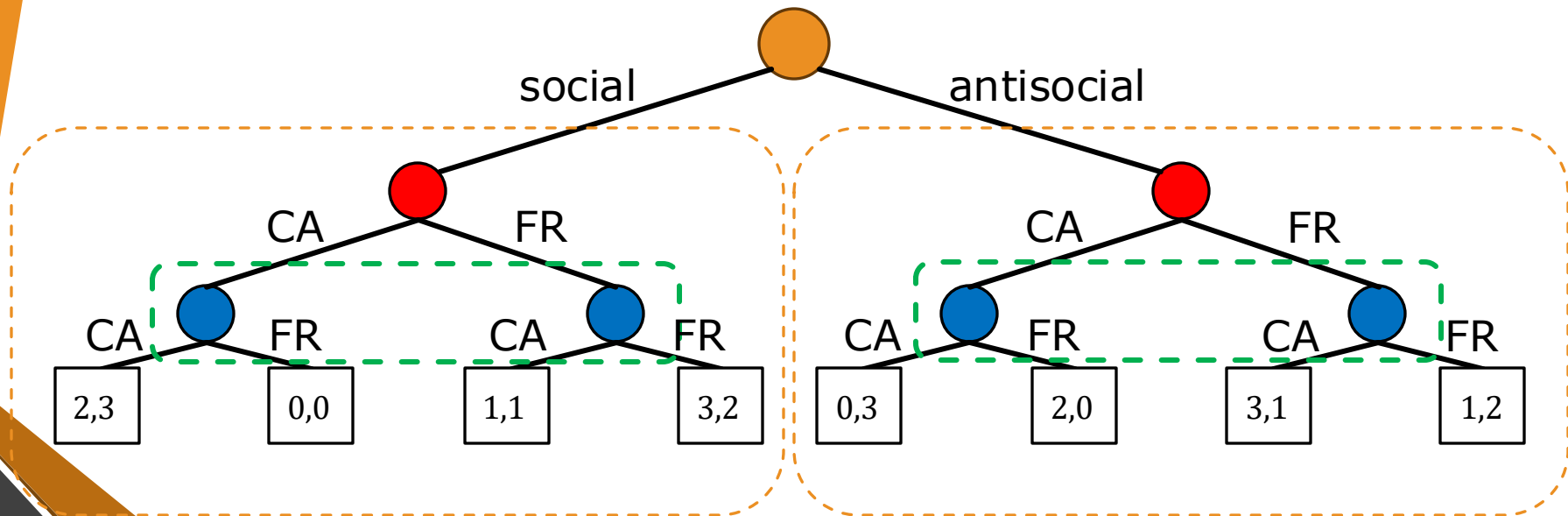
Paula needs to base her action on her prior belief regarding Vignesh's emotional state.



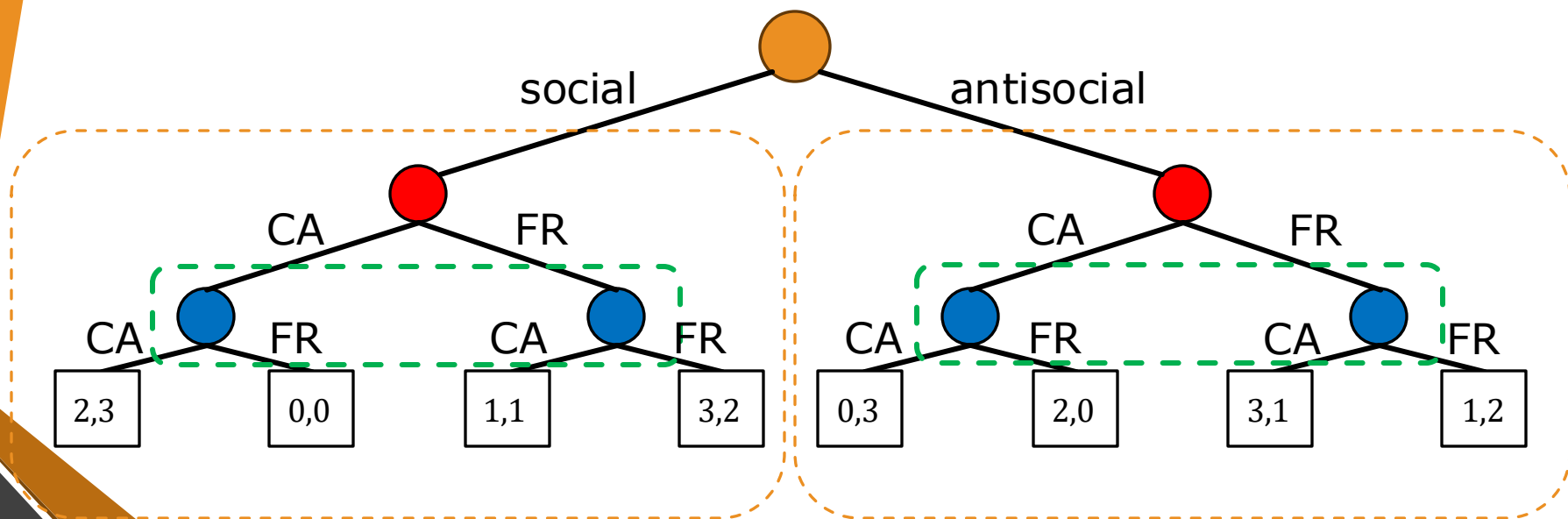
Vignesh's strategy can be of the form (s_1, s_2) where s_1 is his choice if he's feeling social, and s_2 is his choice if he's feeling antisocial.

Paula needs to commit to one choice.

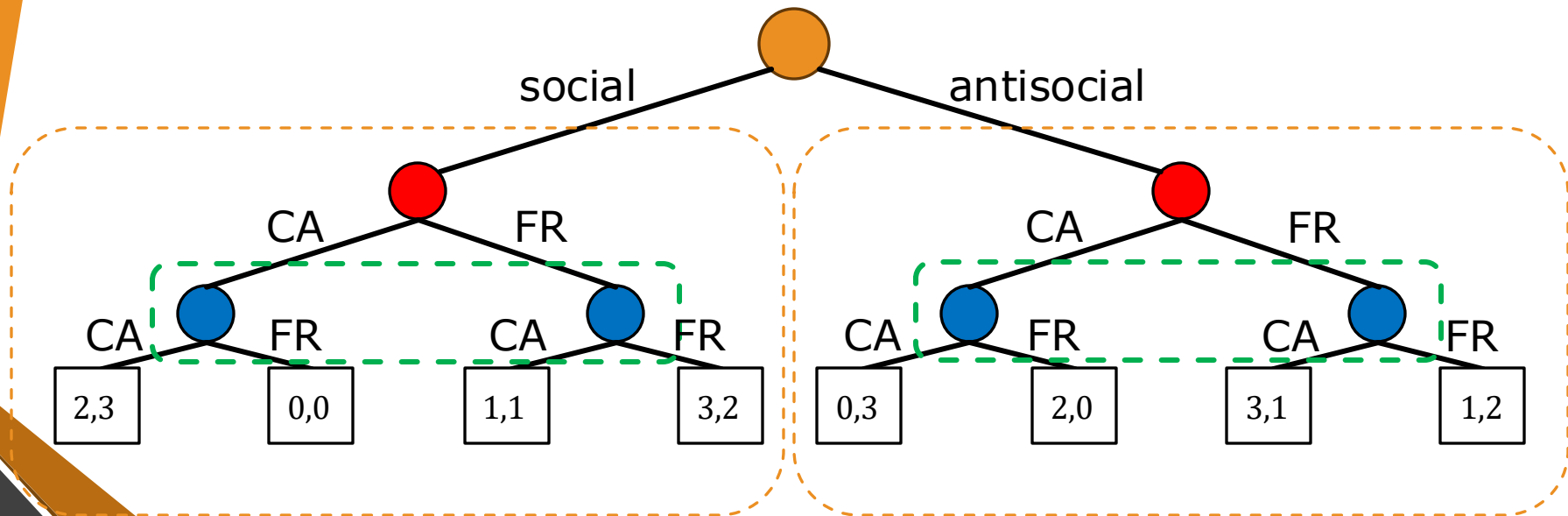
A strategy profile is of the form $((s_1, s_2), q)$



	(CA, CA)	(CA, FR)	(FR, CA)	(FR, FR)
CA	3, (2,0)	2, (2,3)	2, (1,0)	1, (1,3)
FR	0, (0,2)	1, (0,1)	1, (3,2)	2, (3,1)

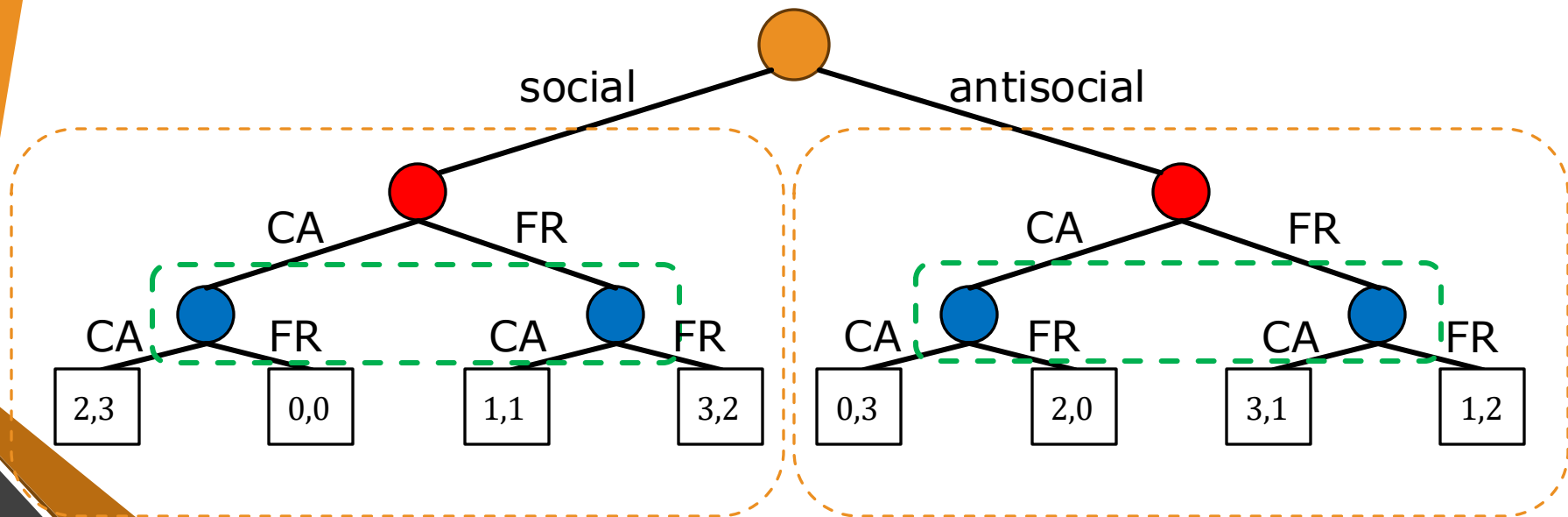


	(CA, CA)	(CA, FR)	(FR, CA)	(FR, FR)
CA	3, (2,0)	2, (2,3)	2, (1,0)	1, (1,3)
FR	0, (0,2)	1, (0,1)	1, (3,2)	2, (3,1)



	(CA, CA)	(CA, FR)	(FR, CA)	(FR, FR)
CA	3, (2,0)	2, (2,3)	2, (1,0)	1, (1,3)
FR	0, (0,2)	1, (0,1)	1, (3,2)	2, (3,1)

Thus, $((CA, FR), CA)$ is a Nash equilibrium (under some definition).



Bayes-Nash Equilibrium

Each player i has a type t_i drawn from a **type space** T_i . Players know their own type, but not other players' types.

Given a strategy profile $\vec{s}$ and a type vector $\vec{t} \in T_1 \times \dots \times T_n$, player i receives a payoff $u_i(\vec{s}, \vec{t}) \in \mathbb{R}$.

Each player has a **belief** about other players' types:

$p_i[\vec{t}_{-i} \mid t_i]$ is the probability that the other players' type vector is $\vec{t}_{-i}$, given that player i 's type is t_i .

The vector $\vec{t}_{-i}$ is in $T_1 \times T_2 \times \dots \times T_{i-1} \times T_{i+1} \times \dots \times T_n$

Thus, given a strategy profile $\vec{s}$ and player i 's type, player i 's expected utility given their type is:

$$u_i(\vec{s} \mid t_i) = \sum_{\vec{t}_{-i}} p_i[\vec{t}_{-i} \mid t_i] \times u_i(\vec{s}, (\vec{t}_{-i}, t_i))$$

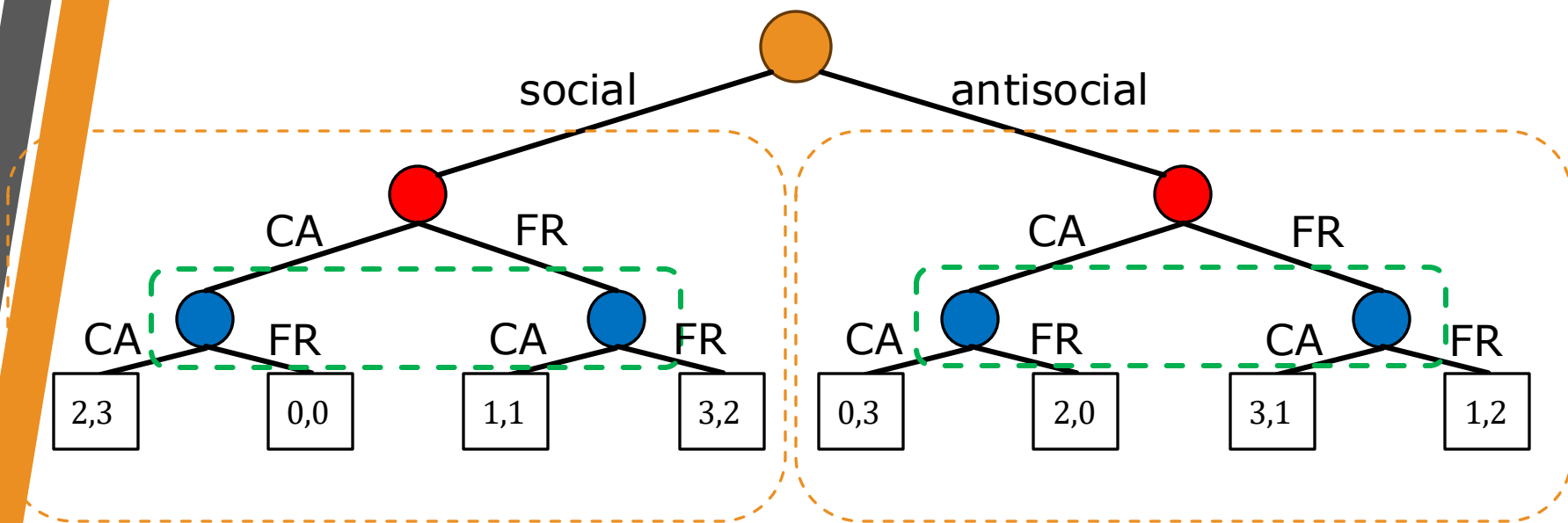
Bayes-Nash Equilibrium

A Bayes-Nash equilibrium is a Nash equilibrium in the game where each player type is a separate player.

A strategy profile $\vec{s}$ is a Bayes-Nash equilibrium if for any type profile $\vec{t}$, any player i and any strategy $d_i \in S_i$

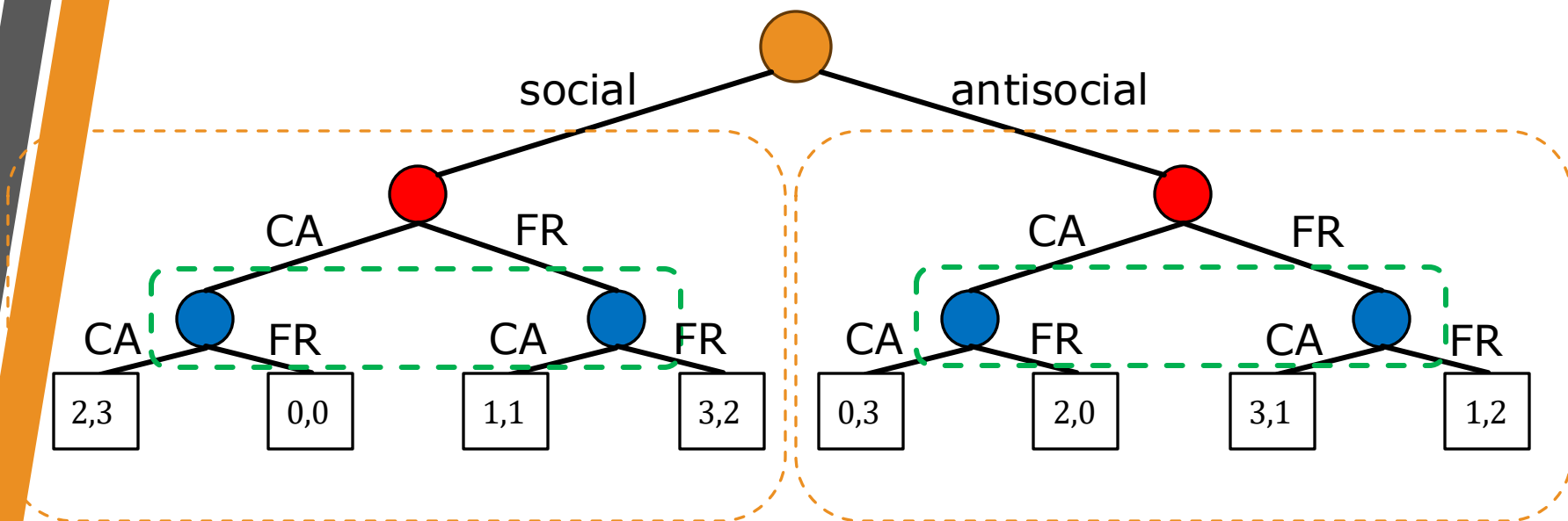
$$u_i(\vec{s} \mid t_i) \geq u_i(\vec{s}_{-i}, d_i \mid t_i)$$

"I prefer to play the strategy s_i over any other strategy d_i given my type t_i ."



Let's change Paula's **prior** to be that Vignesh is social with prob. $\frac{5}{6}$. Her payoffs change:

	(CA, CA)	(CA, FR)	(FR, CA)	(FR, FR)
CA	$\frac{5}{6} \cdot 3 + \frac{1}{6} \cdot 3$	$\frac{5}{6} \cdot 3 + \frac{1}{6} \cdot 1$	$\frac{5}{6} \cdot 1 + \frac{1}{6} \cdot 3$	$\frac{5}{6} \cdot 1 + \frac{1}{6} \cdot 1$
FR	$\frac{5}{6} \cdot 0 + \frac{1}{6} \cdot 0$	$\frac{5}{6} \cdot 0 + \frac{1}{6} \cdot 2$	$\frac{5}{6} \cdot 2 + \frac{1}{6} \cdot 0$	$\frac{5}{6} \cdot 2 + \frac{1}{6} \cdot 2$



Let's change Paula's **prior** to be that Vignesh is social with prob. $\frac{5}{6}$. Her payoffs change:

	(CA, CA)	(CA, FR)	(FR, CA)	(FR, FR)
CA	3, (2,0)	$2\frac{2}{3}, (2,3)$	$1\frac{1}{3}, (1,0)$	1, (1,3)
FR	0, (0,2)	$\frac{1}{3}, (0,1)$	$1\frac{2}{3}, (3,2)$	2, (3,1)

	(CA, CA)	(CA, FR)	(FR, CA)	(FR, FR)
CA	$3, (2,0)$	$2\frac{2}{3}, (2,3)$	$1\frac{1}{3}, (1,0)$	$1, (1,3)$
FR	$0, (0,2)$	$\frac{1}{3}, (0,1)$	$1\frac{2}{3}, (3,2)$	$2, (3,1)$

VS.

	(CA, CA)	(CA, FR)	(FR, CA)	(FR, FR)
CA	$3, (2,0)$	$2, (2,3)$	$2, (1,0)$	$1, (1,3)$
FR	$0, (0,2)$	$1, (0,1)$	$1, (3,2)$	$2, (3,1)$